

Gold–Arf normal play and the observation boundary

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Abstract

This is a consolidated experimental note about the part of `ogdoad` that seems mathematically worth isolating. Conway games under disjunctive sum do not form a scalar ring, so the project does not construct “Clifford algebras over all games”. Instead it builds Clifford algebras over field-like game-value worlds, with emphasis on the characteristic-2 nimber backend, whose engine keeps the quadratic data $q_i = e_i^2$ and the polar data $b_{ij} = e_i e_j + e_j e_i$ independent and computes trace-Arf invariants of quadratic forms over finite nimber subfields.

The experimental bridge is this: the Gold trace forms

$$Q_a(x) = \text{Tr}(x x^{2^a})$$

are standard quadratic forms over fields of characteristic 2, but in the nimber setting they are also composites of combinatorial-game operations: nim-product via Turning Corners, Frobenius squaring via the diagonal product, and XOR via disjunctive sum. Their Arf invariant has the classical zero-count interpretation. The main result of this revision constructs a uniform normal-play game whose P -positions are exactly $\{x : Q_a(x) = 0\}$, so Arf, together with the radical data in the singular case, determines its second-player win-bias.

Beyond the construction, the note synthesizes three threads of work on the formerly open game-semantics question. First, the obstruction landscape: normal-play sums and Winning-Ways coin-turning realize only *linear* P -sets (the lexicode phenomenon is the rich solved case of that linear ceiling), and a family of no-go theorems eliminates symmetric, commutative, oracle-bounded, and translation-equivariant architectures above it. Second, an extraspecial-group reframing: a characteristic-2 quadratic form *is* a central extension $1 \rightarrow \mathbb{Z}/2 \rightarrow E \rightarrow V \rightarrow 1$ whose squaring map is Q , so any group model realizing Q as its squaring map must be noncommutative; this yields a useful vocabulary between the frame-blind no-go and direct global evaluation. Third, the constructive boundary: a public, refinement-blind Witt frame turns the polar support into a matching, while each active original coordinate is a second matched pair whose local overlap has weight q_j . A local FIFO strategy proves that either seat forces zero correction parity on every such board. A one-move terminal charge lantern then compiles the forced bit to ordinary normal play. The rule is refinement-uniform and uses at most one weight-one observation per transition.

That positive theorem is paired with two negative ones. Any transcript-stable exact realizer must observe vectors spanning the input; with weight- w observations, a weight- t input needs at least t/w observations. Thus a constant total observation budget is impossible, and the construction’s singleton source pairs attain the sharp support bound. Conversely, the basic property “there exists a refinement-sensitive winning fork”, even when the fork is required after every completed play, cannot certify non-tautology: an outcome-preserving terminal padding gives every finite normal-play tree that property. The result is an exact dichotomy under declared access disciplines, not a claim that an extensional outcome test can define the word “natural” by itself.

Claim levels. Following the repository convention, statements below are labelled *standard math* (external facts with citations), *implemented and tested* (backed by tests, examples, or experiments in this checkout), *interpretation* (conditional bridges), or *open*. Several results of the 2026-06-10 parallel research run are machine-verified only at stated small parameters; each such result carries its verification scope inline. One earlier construction claim (ECHO-FIFO+dummy, Section 10), recorded as unverified in earlier drafts, has since passed its pre-registered adversarial review (2026-06-10): a fresh direct stateful solver (`experiments/echo_solver.py`) re-derives the full 391,680-check $m = 8$ exactness sweep with zero misses. The verification record is in Section 10. The general theorem proved here is structural rather than extrapolated from that census. Lean kernel-checks the independent all-exponent diagonal-source, matching, adapted-algebra, compiler-equivalence, observation-lower-bound, and fork-padding lemmas used in the synthesis; the formal boundary is stated precisely in Section 13.

1 Scope

A short partizan game is built recursively as $\{L \mid R\}$, and games form a partially ordered abelian group under disjunctive sum [4, 3]. They do not form a ring: Conway multiplication is defined on the number subclass, not on arbitrary games. Since a Clifford algebra needs a commutative scalar ring, the phrase “Clifford over games” has to be interpreted through scalar subworlds:

world	algebraic structure	role here
No	real-closed field, char 0	ordinary real Clifford theory with surreal scalars
No [i]	algebraically closed field, char 0	ordinary complex Clifford theory
finite nimber subfields	finite fields of char 2	characteristic-2 Clifford and Arf experiments

The implemented `Nimber` scalar is finite: `u128` realizes $\mathbb{F}_{2^{128}}$, containing the nimber subfields $\mathbb{F}_{2^m}$ for $m = 1, 2, 4, \dots, 128$. It is not the whole proper-class field On_2 .

2 The char-2 Clifford datum

In characteristic 2, a quadratic form is not determined by its polar form. For basis vectors the Clifford relations are

$$e_i^2 = q_i, \quad e_i e_j + e_j e_i = b_{ij}.$$

The polar form is alternating, so $b_{ii} = 0$, but q_i can be nonzero. A faithful characteristic-2 implementation must store q and b separately. This is the main implementation discipline behind the nimber backend, and – as Section 7 makes precise – it is also the group-theoretic content of the whole note: q and b are the independent central data (squares and commutators) of an extraspecial-type extension.

For a nonsingular quadratic form over $\mathbb{F}_2$ with symplectic basis (a_k, b_k) , the Arf invariant is

$$\text{Arf}(Q) = \sum_k Q(a_k)Q(b_k) \in \mathbb{F}_2.$$

It classifies nonsingular binary quadratic spaces at fixed dimension. Over a finite characteristic-2 field, the corresponding value lies in $F/\wp(F)$, with $\wp(y) = y^2 + y$; for finite fields this quotient is read by the field trace to $\mathbb{F}_2$.

Remark 1. Ovsienko [11] identifies a precise bridge between Dickson’s classification of quadratic forms over $\mathbb{F}_2$ and the classical classification of real Clifford algebras. We use that paper as motivation for the Arf/Clifford connection, not as a blanket classification theorem for all Clifford algebras over finite number fields. In the code, rank and radical data are reported alongside Arf, and degenerate forms are not collapsed to a single Arf bit.

The small finite-field check used throughout is:

$$q = (*2, *2), \quad b_{01} = *1 \quad \text{has Arf } 1,$$

whereas

$$q = (*2, *3), \quad b_{01} = *1 \quad \text{has Arf } 0.$$

The polar entry is part of the form.

3 Gold forms from game operations

Nim-addition is XOR, and XOR is the disjunctive sum of impartial game values. Nim-multiplication is also game-theoretic: Conway’s Turning-Corners product has the mex recurrence

$$x \otimes y = \text{mex} \{ (i \otimes y) \oplus (x \otimes j) \oplus (i \otimes j) : 0 \leq i < x, \ 0 \leq j < y \}. \quad (1)$$

The repo implements this recurrence directly for small arguments and checks it against the fast algebraic nim product.

Thus the following operations are game-realizable at the level of number values:

$$x \oplus y, \quad x \otimes y, \quad x \mapsto x^2 = x \otimes x, \quad \text{Tr}(x) = x + x^2 + \cdots + x^{2^{m-1}}.$$

For $F = \mathbb{F}_{2^m}$ and $a \geq 1$, define the Gold trace form

$$Q_a(x) = \text{Tr}(x x^{2^a}) = \text{Tr}(x^{1+2^a}).$$

Since Frobenius $x \mapsto x^{2^a}$ is additive, Q_a is a quadratic form over $\mathbb{F}_2$. In the nimber backend, it is also a composite of the game operations above. Throughout, “the instance (m, a) ” means Q_a on $\mathbb{F}_{2^m}$, and “ (m, a, λ) ” the scaled component $Q_{\lambda, a}(x) = \text{Tr}(\lambda x^{1+2^a})$.

Observation 1. *The Gold forms are not merely defined on a field of game values. In the finite nimber fields tested here, they are built from nim/game operations: Turning Corners for multiplication, diagonal Turning Corners for Frobenius squaring, and disjunctive sum for XOR and trace.*

This observation is modest but useful: it gives a concrete quadratic form at the interface between impartial-game arithmetic and characteristic-2 quadratic-form theory.

field	a	Q	Arf	type	rank	formula
$\mathbb{F}_{2^4}$	1	$\text{Tr}(x^3)$	1	O^-	2	2
$\mathbb{F}_{2^8}$	1	$\text{Tr}(x^3)$	1	O^-	6	6
$\mathbb{F}_{2^8}$	2	$\text{Tr}(x^5)$	1	O^-	4	4
$\mathbb{F}_{2^{16}}$	1	$\text{Tr}(x^3)$	1	O^-	14	14
$\mathbb{F}_{2^{16}}$	4	$\text{Tr}(x^{17})$	1	O^-	8	8
$\mathbb{F}_{2^{32}}$	1	$\text{Tr}(x^3)$	1	O^-	30	30

Table 1: Representative output of `experiments/trace_form_arf.py`. The script checks all 15 cases with $m = 2, 4, 8, 16, 32$ and $1 \leq a \leq \log_2 m$.

3.1 Rank check

The polar rank of Gold forms is known. In the tested regime $m = 2^k$ and $1 \leq a \leq k$, the general formula

$$\text{rank } B_{Q_a} = m - \gcd(2a, m)$$

specializes to $m - 2 \gcd(a, m)$ because $m / \gcd(a, m)$ is even. The classifier reproduces the general formula in all tested cases. Representative rows are shown in Table 1.

Most of these forms have a nontrivial radical. The statement is therefore not “nondegenerate Gold form” but “positive-rank quadratic form with a nonsingular core whose Arf value is computed after symplectic reduction.”

3.2 A broader trace family

The coefficient-1 Gold form is only one trace monomial. A standard trace description of the quadratic part of Boolean functions over $\mathbb{F}_{2^m}$ uses terms

$$\text{Tr}(c_i x^{1+2^i})$$

plus affine terms and, in even degree, a middle half-trace term. The monomials displayed here are still game-operation-built in the nimber backend: $c_i x^{1+2^i} = c_i \otimes x \otimes x^{2^i}$, followed by trace/XOR.

The current implementation does not claim to package every Boolean quadratic function. The probe `experiments/gold_family_survey.py` deliberately omits the middle term and affine book-keeping. What it does check, exhaustively over $\mathbb{F}_{256}$, is that scaled Gold components

$$Q_{\lambda,a}(x) = \text{Tr}(\lambda x^{1+2^a})$$

already reach nondegenerate (bent) forms for many λ . For APN Gold exponents in that field, the observed count matches the classical $2(2^m - 1)/3$ bent-component count. This matters for the game question because a bent form has trivial radical: the loopy radical route collapses to $\{0\}$, and the frame-blind symplectic no-go applies without a degenerate layer to hide in.

Remark 2 (a vacuous small target). At $m = 4$, $a = 1$ the zero set $\{Q_1 = 0\} \subset \mathbb{F}_{16}$ is the subfield $\mathbb{F}_4$, an $\mathbb{F}_2$ -subspace of dimension 2: the form is genuinely quadratic (polar rank 2), but its zero set is affine, so as a target for “is this P -set a genuine quadric” it is vacuous – a linear rule realizes it. The smallest honest unscaled instance is $(8, 1)$. The repository’s quadric-fitting layer (`src/forms/quadric_fit.rs`) reports exactly this distinction (genuine polar rank versus affine flat), and every probe below that consumes a fitted quadric is read with that guard in mind.

4 Arf as win-bias

The relevant counting theorem is classical. If Q is nonsingular on $\mathbb{F}_2^{2r}$, then

$$\#\{x : Q(x) = 0\} = 2^{2r-1} + (-1)^{\text{Arf}(Q)} 2^{r-1}.$$

For degenerate forms, the code uses the standard radical-adjusted count: an anisotropic radical balances the values, while an isotropic radical scales the bias of the nonsingular core.

The counting theorem becomes the game interpretation after Theorem 8:

For weighted-source WITT-FIFO, P -positions are exactly $\{x : Q(x) = 0\}$. On a non-singular core the Arf invariant is the sign of the bias between second-player wins and first-player wins; in the singular case the radical data supplies the standard adjustment.

Historically this statement was only the target that motivated the search; Section 11 supplies the rule and proof.

field	a	Arf	rank	$\#\{Q = 0\}$	bias
$\mathbb{F}_{2^4}$	1	1	2	4	-4
$\mathbb{F}_{2^8}$	1	1	6	112	-16
$\mathbb{F}_{2^8}$	2	1	4	96	-32
$\mathbb{F}_{2^{16}}$	1	1	14	32512	-256
$\mathbb{F}_{2^{16}}$	4	1	8	30720	-2048

Table 2: Representative zero-counts from `experiments/arf_win_bias.py`. The bias is $\#\{Q = 0\} - 2^{m-1}$.

5 The linear ceiling

For a normal-play disjunctive sum of impartial games, the P-condition is linear in Grundy coordinates:

$$g_1 \oplus \cdots \oplus g_n = 0.$$

So normal-play sums give linear P-sets, not genuine quadrics. The coin-turning architecture – the very architecture whose product theorem defines the nim product – is subject to the same ceiling.

Theorem A (coin-turning linearity; standard math, machine cross-checked). *Every Winning Ways coin-turning game has configuration P-set equal to the kernel of an $\mathbb{F}_2$ -linear nimber-valued map. In particular no coin-turning P-set is a quadric of positive polar rank.*

Proof sketch. A coin-turning position is a finite set of heads; by the Winning Ways coin-turning theory [3], the Grundy value of a configuration is the XOR of the single-coin Grundy values $g(i)$ of its heads. The P -set is therefore $\{x : \bigoplus_{i \in x} g(i) = 0\}$, the kernel of the $\mathbb{F}_2$ -linear map $x \mapsto \bigoplus_i x_i g(i)$. The repository cross-check (`experiments/gold/octal_attack.py`) verifies XOR-additivity by mex recursion, exhaustively over all 32,768 companion-family choices on 4 coins. $\square$

Theorem A has a sharp classical complement: the linear ceiling is exactly where coding theory already harvests games.

Remark 3 (the lexicode shadow; standard math and interpretation). Conway–Sloane lexicodes [5] are built by the greedy lexicographic rule – the mex rule – and the 1986 paper realizes them as the Grundy-value-0 positions of associated turning-game move structures; binary lexicodes are linear *because of* Sprague–Grundy theory (XOR-closure is a game theorem there, not a coding theorem). They include the Hamming codes and the length-24, $d = 8$ extended binary Golay code; over base 2^k they are closed under nim-addition, and they are linear at the Fermat-power bases 2^{2^k} – exactly the sizes at which nim-multiplication makes the ordinals below the base a field. Read together with Theorem A: lexicodes are the *floor* (rich linear codes naturally realized as Grundy-0 sets) and Theorem A is the *ceiling* (coin-turning realizes only linear P -sets); floor and ceiling meet at linear. The former open problem was whether the lexicode phenomenon admits a quadratic refinement. The obstruction is precisely the XOR-closure failure that the polar form measures, and weighted-source FIFO resolves it outside disjunctive sums.

For a quadratic form in characteristic 2,

$$Q(u + v) = Q(u) + Q(v) + B(u, v).$$

Thus for $u, v \in \{Q = 0\}$,

$$u + v \in \{Q = 0\} \iff B(u, v) = 0.$$

The polar form is exactly the obstruction to XOR-closure. The first two layers are game-built: XOR is disjunctive sum and B is built from nim products. The play layer supplied in Section 11 integrates that bilinear data into the quadratic zero set.

6 Routes and obstructions

6.1 Test benches

The repo contains several probes. They are useful for narrowing the target, but they are not a proof that no natural game exists.

- `fit_f2_quadratic` tests whether a subset of $\mathbb{F}_2^k$ is the zero set of a quadratic polynomial, and whether the result has nonzero polar rank (distinguishing genuine quadrics from affine flats; cf. Remark 2).
- A bounded misere quotient computer tests small quotient P -sets. The octal sweep checked 292 octal codes, heap cutoff at most 4, and found quotient orders 2, 6, 10, 12, 14, with no elementary abelian 2-group of rank at least 2. Theorem C below shows this emptiness is forced, not a sampling accident.
- `experiments/misere_kernel.py` checks the kernel obstruction on the smallest wild misere quotient R_8 : the canonical group/kernel shadow is linear, while the genuinely misere P -element lies outside that vector-space part. Corollary 1 below shows the linearity is forced by commutativity.
- An interactive-kernel solver tests finite move graphs. It confirms that an ad hoc acyclic game can realize any subset, and that a rule referencing Q itself realizes $\{Q = 0\}$ tautologically. The tested B -coupled rules do not produce the Gold zero set.
- `examples/loopy_quadric.rs` tests cyclic move graphs with Draw sets. The symmetric B -coupled rule has Loss-set equal to the radical $R(B)$, so it reproduces the obstruction rather than the Gold zero set; this is the special case of Theorem G below, and its small (4, 1) “hit” is the radical coincidence $|\{Q_1 = 0\}| = 4 = |R(B)|$ there.

- `examples/bent_route.rs` repeats route probes on a bent scaled Gold component. In that example, a B plus coordinate-frame rule reaches a quadric of the correct Arf class but not the specific Gold zero set; the naive local-field/spin-flip completion leaves the quadric variety.

Except for the frame-blind proposition below and the lettered theorems of Section 8, these are bounded or example-level negative results.

6.2 Frame-blind no-go

There is a clean obstruction for rules that see only a nondegenerate symplectic form and no frame.

Proposition 1. *Let B be a nondegenerate alternating form on $V = \mathbb{F}_2^{2r}$ with $r \geq 2$. If a finite normal-play game has position set V and a move relation invariant under $\mathrm{Sp}(B)$, then its P -set is not a nondegenerate quadric.*

Proof. Each element of $\mathrm{Sp}(B)$ is an automorphism of the move graph, so the retrograde outcome function is $\mathrm{Sp}(B)$ -invariant. The P -set is therefore a union of symplectic orbits. For $r \geq 2$, the symplectic group is transitive on $V \setminus \{0\}$ [14], so the invariant subsets are only $\emptyset$, $\{0\}$, $V \setminus \{0\}$, and V . A nondegenerate quadric in dimension at least 4 has $2^{2r-1} \pm 2^{r-1}$ points, which is none of these. $\square$

The hypothesis matters. Coordinate-dependent rules have already broken $\mathrm{Sp}(B)$ symmetry, so this proposition does not explain every negative probe. It explains only genuinely frame-blind B -only rules. Section 8 extends the obstruction far beyond this symmetric class.

6.3 The diagonal framing

Once a coordinate frame is allowed, a canonical-looking quadratic refinement of B is

$$Q_{\text{frame}}(v) = \sum_{i < j} B(e_i, e_j) v_i v_j.$$

For the Gold polar forms tested in this repo, the Gold form decomposes as

$$Q_a(v) = Q_{\text{frame}}(v) + \ell_{\text{diag}}(v), \quad \ell_{\text{diag}}(v) = \sum_i Q_a(e_i) v_i.$$

In the tested genuinely quadratic Gold cases up to $\mathbb{F}_{2^{16}}$, Q_{frame} has Arf 0 and the diagonal term changes the form to the Gold Arf-1 class.

Remark 4. This is an experimental pattern for the Gold polar forms tested here, not a theorem about every alternating form plus every coordinate frame. Random nondegenerate alternating forms can give a zero-diagonal frame quadric of Arf 1.

The diagonal $q_i = Q_a(e_i)$ is therefore the named missing datum. Two results of the parallel run locate it more precisely.

Proposition 2 (game-native diagonal source for $a = 1$; implemented and tested). *Let $m = 2^k$ and let $w \in \mathbb{F}_{2^m}$ be the XOR of the Fermat coins 2^{2^t} for $t = 1, \dots, k-1$. Then the $a = 1$ Gold diagonal satisfies*

$$q_i^{(m,1)} = Q_1(e_i) = \mathrm{Tr}(\wp(w) e_i), \quad \wp(w) = w \otimes w \oplus w,$$

i.e. the trace-dual element of the diagonal is $\wp(w)$. Verified at $m = 4, 8, 16, 32$ via the trace-pairing uniqueness of the dual element (`experiments/gold/gold_diag_probe.py`).

This gives a game-native, tower-uniform, commutative source for the $a = 1$ diagonal: one nim-squaring, one XOR, one trace – all inside the licensed operation chain of Section 3. The following theorem closes the same sourcing question for every exponent, including even a .

Theorem 1 (all-exponent diagonal source). *Let $m = 2^k \geq 2$ and let $\lambda_{a,c}^{(m)}$ be the unique trace-dual of the scaled diagonal*

$$\mathrm{Tr}_m(\lambda_{a,c}^{(m)} e_i) = \mathrm{Tr}_m(c e_i \mathrm{Fr}^a(e_i)).$$

For $m = 2M$, write $\mathbb{F}_{2^m} = \mathbb{F}_{2^M}(u)$ with $u = e_M$ and $u^{2^M} = u + 1$. Put

$$d_0 = \mathrm{Tr}_{m/M}(c), \quad d_1 = \mathrm{Tr}_{m/M}(cu \mathrm{Fr}^a(u)), \quad L_j = \lambda_{a,d_j}^{(M)}.$$

Then

$$\lambda_{a,c}^{(2M)} = (L_0 + L_1) + uL_0.$$

In particular, for $c = 1$ one has $L_0 = 0$, so $\lambda_{a,1}^{(2M)} \in \mathbb{F}_{2^M}$ and $\mathrm{Tr}_m(\lambda_{a,1}^{(2M)}) = 0$. Consequently there is a $w \in \mathbb{F}_{2^m}$ with

$$\lambda_{a,1}^{(m)} = w^2 + w, \quad Q_a(e_i) = \mathrm{Tr}_m((w^2 + w)e_i)$$

for every i and every a .

The proof is the quadratic-tower trace calculation in `writeups/gold_diagonal_source.tex`. The recursion is implemented by `gold_component_diagonal_dual`; `gold_diagonal_artin_schreier_source` then solves the final Artin–Schreier equation. The tower block identities and the absolute-trace descent are kernel-checked in `formal/Ogdoad/GoldDiagonal.lean`. Thus the previously observed drift records the recursive dual rather than an obstruction: every unscaled dual has a uniform constructive preimage. The claim is deliberately unscaled; a general coefficient c need not descend or have trace zero. The source w is selected by exact finite-field linear algebra, not by a simple Fermat-coin formula as in Proposition 2; this is a uniform constructive source, not yet a strategic naturality result.

Remark 5 (subfield vanishing; standard math). $q_i^{(m,a)} = 0$ for all $i < m/2$: the Fermat basis elements at lower indices, their Frobenius iterates, and their nim products stay inside the half-field, whose absolute trace from an even-degree extension vanishes. The Gold diagonal lives on the top Fermat layer only.

Remark 6 (resolved diagonal-framing question). The diagonal framing

$$q_i = Q_a(e_i) = \mathrm{Tr}(e_i^{1+2^a})$$

is game-built for every a : Proposition 2 gives the explicit $a = 1$ source, and Theorem 1 gives a constructive source for every exponent. The weighted-source Witt–FIFO theorem in Section 11 supplies the formerly missing local normal-play rule with P-set $\{x : Q_a(x) = 0\}$.

7 The extraspecial reframing

The diagonal-framing question can be restated in group-theoretic terms, and the restatement sharpens it. The dictionary is classical – the characteristic-2 Heisenberg/extraspecial picture of quadratic forms [13, 9] – and this section proves the pieces we use, to keep the note self-contained. Claim levels: Lemmas 1, 2, 3 and Proposition 3 are standard mathematics; the R_8 instantiation in Corollary 1 is implemented and tested (`experiments/misere_kernel.py`); reading E -equivariance as

a naturality criterion is interpretation. The weighted-source theorem below realizes the quadric without claiming that ordinary play itself is a model of the central extension E ; that stricter extension-model question is optional.

Throughout, $V = \mathbb{F}_2^n$ written additively, and $Z = \{1, z\} \cong \mathbb{Z}/2$ written multiplicatively. A quadratic map $Q: V \rightarrow \mathbb{F}_2$ is a function with $Q(0) = 0$ whose polarization $B(x, y) = Q(x + y) + Q(x) + Q(y)$ is bilinear; B is then alternating (hence symmetric), and Q may have any diagonal $q_i = Q(e_i)$, exactly the char-2 datum the engine keeps separate from b .

7.1 Quadratic forms are central extensions

Lemma 1 (extraspecial dictionary). *Let $1 \rightarrow Z \rightarrow E \xrightarrow{\pi} V \rightarrow 1$ be a central extension of the elementary abelian 2-group V by $Z \cong \mathbb{Z}/2$.*

- (i) *For $x \in V$ and any lift $\tilde{x} \in \pi^{-1}(x)$ one has $\tilde{x}^2 \in Z$, and $\tilde{x}^2$ is independent of the lift; the squaring form $Q_E: V \rightarrow \mathbb{F}_2$ defined by $\tilde{x}^2 = z^{Q_E(x)}$ is well defined, as is the commutator pairing B_E defined by $[\tilde{x}, \tilde{y}] = z^{B_E(x, y)}$, which is bilinear and alternating. They satisfy*

$$(\tilde{x}\tilde{y})^2 = \tilde{x}^2 \tilde{y}^2 [\tilde{x}, \tilde{y}], \quad \text{equivalently} \quad Q_E(x + y) = Q_E(x) + Q_E(y) + B_E(x, y),$$

so Q_E is a quadratic map with polar form B_E .

- (ii) *Conversely, every quadratic map Q arises: fixing a basis $e_1, \dots, e_n$ of V , the bilinear 2-cocycle*

$$\varphi(x, y) = \sum_i x_i y_i Q(e_i) + \sum_{i > j} x_i y_j B(e_i, e_j)$$

defines a group $E_Q = Z \times V$ with $(s, x)(t, y) = (s + t + \varphi(x, y), x + y)$, whose squaring form is Q and whose commutator pairing is the polar form B .

- (iii) *The assignment $Q \mapsto [E_Q]$ is a bijection from quadratic maps on V to equivalence classes of central extensions of V by $\mathbb{Z}/2$.*

- (iv) *E_Q is abelian iff $B = 0$ (iff Q is linear); and for $n \geq 1$, E_Q is extraspecial (i.e. $Z(E) = [E, E] = \Phi(E) \cong \mathbb{Z}/2$) iff B is nondegenerate, which forces $n = 2r$ even and $|E_Q| = 2^{1+2r}$.*

Proof. (i) $\pi(\tilde{x}^2) = 2x = 0$, so $\tilde{x}^2 \in Z$; replacing $\tilde{x}$ by $z\tilde{x}$ multiplies the square by $z^2 = 1$. Commutators of lifts lie in Z because V is abelian, are central, and are unchanged by central retagging of the lifts; centrality of commutators gives bimultiplicativity, and $[\tilde{x}, \tilde{x}] = 1$ gives $B_E(x, x) = 0$. For the identity: with $c = [\tilde{y}, \tilde{x}]$ central, $\tilde{x}\tilde{y}\tilde{x}\tilde{y} = \tilde{x}(\tilde{x}\tilde{y}c)\tilde{y} = \tilde{x}^2\tilde{y}^2c$, and $c = c^{-1} = [\tilde{x}, \tilde{y}]$ since Z has exponent 2. As $\tilde{x}\tilde{y}$ lifts $x + y$, the additive identity follows.

(ii) A bilinear map is a 2-cocycle, so E_Q is a group with center containing $Z = \{(0, 0), (1, 0)\}$. Squares: $(s, x)^2 = (\varphi(x, x), 0)$ and $\varphi(x, x) = \sum_i x_i Q(e_i) + \sum_{i > j} x_i x_j B(e_i, e_j) = Q(x)$ by the polarization expansion of Q in the basis. Commutators: $(s, x)(t, y)(s, x)^{-1}(t, y)^{-1} = (\varphi(x, y) + \varphi(y, x), 0) = (B(x, y), 0)$, using that the symmetrization of φ is B (the diagonal terms cancel and B is symmetric).

(iii) Equivalent extensions have cohomologous cocycles, and a coboundary $\delta\psi(x, y) = \psi(x) + \psi(y) + \psi(x + y)$ has zero diagonal ($\delta\psi(x, x) = \psi(0) = 0$), so the squaring form is an invariant of the class; (ii) gives surjectivity. Injectivity: if two extensions have equal squaring form Q , choose sections and cocycles φ, φ' ; both have diagonal Q , and both have symmetrization equal to the polar form of Q , so $d = \varphi + \varphi'$ is a symmetric cocycle with zero diagonal. The extension E_d is then abelian

of exponent 2, i.e. an $\mathbb{F}_2$ -vector space, so $1 \rightarrow Z \rightarrow E_d \rightarrow V \rightarrow 1$ splits and d is a coboundary; the two extensions are equivalent.

(iv) Commutators generate $z^{\text{im } B}$, so E_Q is abelian iff $B = 0$. In general $Z(E_Q) = \pi^{-1}(\text{rad } B)$, since $\tilde{x}$ is central iff $B(x, \cdot) = 0$; thus $Z(E_Q) = Z$ iff B is nondegenerate. If so, $B \neq 0$ gives $[E, E] = Z$, and $\Phi(E) = E^2[E, E] = Z$ because all squares lie in Z and E/Z is elementary abelian. A nondegenerate alternating form exists only in even dimension. $\square$

Remark 7. This is the characteristic-2 Heisenberg picture: E_Q is the Heisenberg group of (V, B) and Q selects the central extension; cohomologically, (iii) is the standard identification of $H^2(V; \mathbb{F}_2)$ with quadratic maps $V \rightarrow \mathbb{F}_2$ [13]. The same finite-quadratic-module data drives the Weil S/T matrices in the library's integral layer. Note that the dictionary is exactly the engine's discipline in group clothing: the diagonal q_i (squares) and the off-diagonal b_{ij} (commutators) are independent central data, and collapsing them is collapsing E_Q onto an abelian quotient.

7.2 Arf classifies the two extraspecial types

Write H for the hyperbolic plane ($Q(e_1) = Q(e_2) = 0$, $B(e_1, e_2) = 1$; $\text{Arf} = 0$) and A for the anisotropic plane ($Q \equiv 1$ on $V \setminus \{0\}$; $\text{Arf} = 1$), and $\circ$ for the central product (identify the centers).

Lemma 2 (Arf = the two types). *Let B be nondegenerate, $\dim V = 2r \geq 2$.*

- (i) $E_{Q \perp Q'} \cong E_Q \circ E_{Q'}$.
- (ii) $E_H \cong D_8$ and $E_A \cong Q_8$.
- (iii) Every extraspecial 2-group of order 2^{1+2r} is isomorphic to E_Q for a nondegenerate Q , and $E_Q \cong E_{Q'}$ iff $(V, Q) \cong (V', Q')$ iff the ranks and Arf invariants agree [6, 1]. Hence there are exactly two extraspecial groups of order 2^{1+2r} :

$$E_{2^{1+2r}}^+ = D_8^{\circ r} \text{ (Arf} = 0, Q \cong H^{\perp r}), \quad E_{2^{1+2r}}^- = D_8^{\circ(r-1)} \circ Q_8 \text{ (Arf} = 1, Q \cong H^{\perp(r-1)} \perp A).$$

In particular $D_8 \circ D_8 \cong Q_8 \circ Q_8$, the group avatar of $H \perp H \cong A \perp A$ (additivity of Arf).

- (iv) (Census.) With $\varepsilon = (-1)^{\text{Arf } Q}$,

$$\#\{g \in E_Q : g^2 = 1\} = 2 \#\{x : Q(x) = 0\} = 2^{2r} + \varepsilon 2^r,$$

so the element-order census of E_Q is the zero-count bias of Section 4 read multiplicatively, and it distinguishes the two types.

Proof. (i) The cocycle $\varphi \oplus \varphi'$ on $V \oplus V'$ is bilinear with diagonal $Q \perp Q'$, and $(s, (x, x')) \mapsto [((s, x), (0, x'))]$ is an isomorphism onto $(E_Q \times E_{Q'}) / \langle (z, z') \rangle$.

(ii) E_H has order 8, is nonabelian ($B \neq 0$), and has a noncentral involution (any lift of e_1 squares to $z^{Q(e_1)} = 1$), so $E_H \cong D_8$; in E_A every noncentral element squares to z (as $Q \equiv 1$ off 0), so z is the unique involution and $E_A \cong Q_8$.

(iii) For E extraspecial, $V = E/Z(E)$ is elementary abelian (as $\Phi(E) = Z(E)$), so Lemma 1 applies to $1 \rightarrow Z(E) \rightarrow E \rightarrow V \rightarrow 1$ and gives $E \cong E_{Q_E}$ with B_{Q_E} nondegenerate ($Z(E)$ exactly central). If $\psi: E_Q \rightarrow E_{Q'}$ is an isomorphism, it preserves the (characteristic) centers, fixes $z \mapsto z'$ (the unique nontrivial central element), and induces an $\mathbb{F}_2$ -linear map $\bar{\psi}: V \rightarrow V'$ with $Q'(\bar{\psi}x)$ read from $\psi(\tilde{x})^2 = \psi(\tilde{x}^2)$ – an isometry. Conversely an isometry g transports the cocycle: $\varphi' \circ (g \times g)$ has diagonal $Q' \circ g = Q$, hence by Lemma 1(iii) yields an extension equivalent to E_Q ,

and $(s, x) \mapsto (s, gx)$ closes the isomorphism. Dickson's classification of nonsingular quadratic forms over $\mathbb{F}_2$ (two isometry classes per even rank, separated by Arf) finishes; the central-product normal forms follow from (i), (ii) and Witt decomposition.

(iv) The two lifts of x both square to $z^{Q(x)}$, so they are involutions or the identity iff $Q(x) = 0$; now apply the zero-count formula of Section 4. The counts differ for $\varepsilon = \pm 1$, so $E^+ \not\cong E^-$. $\square$

7.3 The abelian obstruction

Lemma 3 (abelian obstruction). *Let M be a commutative monoid, $Z = \{1, z\} \subseteq M$ a two-element subgroup, $N \subseteq M$ a submonoid containing Z , and $\pi: N \twoheadrightarrow V$ a surjective monoid homomorphism onto an $\mathbb{F}_2$ -vector space with $\pi^{-1}(\pi(m)) = mZ$ for all $m \in N$. Then the squaring form $Q: V \rightarrow \mathbb{F}_2$, defined by $\tilde{m}^2 = z^{Q(x)}$ for any $\tilde{m} \in \pi^{-1}(x)$, is well defined and $\mathbb{F}_2$ -linear; its polar form vanishes identically. Consequently no commutative monoid realizes, through its own squaring map, a quadratic form with nonzero – a fortiori nondegenerate – characteristic-2 polar form. Equivalently, by Lemma 1(iv): if $B \neq 0$, the extension E_Q is nonabelian and admits no model (N, Z, π) inside any commutative monoid.*

Proof. $\pi(m^2) = 2\pi(m) = 0$, so $m^2 \in \pi^{-1}(0) = Z$, and $(zm)^2 = z^2m^2 = m^2$, so Q is well defined. Commutativity gives $(mn)^2 = m^2n^2$; since $\tilde{m}\tilde{n}$ is a preimage of $x+y$, this is $Q(x+y) = Q(x)+Q(y)$, i.e. $B \equiv 0$. $\square$

Corollary 1 (misère quotients: the kernel shadow is linear). *Let $\mathcal{Q}$ be a finite misère quotient – a finite commutative monoid – with kernel K , the maximal subgroup of $\mathcal{Q}$ (the mutual-divisibility class of the product of all idempotents) [12].*

- (i) *Every configuration (N, Z, π) as in Lemma 3 inside $\mathcal{Q}$ has linear squaring form: the intrinsic multiplication of a misère quotient cannot supply a quadratic refinement with $B \neq 0$, on K or anywhere else in $\mathcal{Q}$.*
- (ii) *Under the regularity hypothesis of [12, Thm. 6.4] (satisfied by the regular finite quotients arising in practice), $K \cong (\mathbb{Z}/2)^k$ is the group of normal-play Grundy values and the P -portion meets K in the XOR-linear normal-play set.*
- (iii) *On the smallest wild quotient $R_8 = \langle a, b, c \mid a^2 = 1, b^3 = b, bc = ab, c^2 = b^2 \rangle$ with $P = \{a, b^2\}$: the idempotents are $\{1, b^2\}$, the kernel is $K = \{b^2, b, ab, ab^2\} \cong (\mathbb{Z}/2)^2$ with identity b^2 , $P \cap K = \{b^2\} \mapsto \{0\}$ is linear, and the genuinely misère P -element a lies outside K . (Implemented and tested: `experiments/misere_kernel.py`.)*

So the linear obstruction observed on R_8 is forced by commutativity, not an accident of the example: a nondegenerate polar form is the commutator pairing of a nonabelian group, and a misère quotient has none to offer.

Proof. (i) is Lemma 3 applied to submonoids of $\mathcal{Q}$, which are commutative. (ii) is quoted from [12]. (iii) is a finite verification. $\square$

7.4 The integral shadow

The same extension data has an integral-lattice avatar, already shipped in the library (interpretation, verified through the `DiscriminantForm + arf_number` pipeline): the Gold nonsingular core of rank $2r$ and Arf ε is the discriminant form of the even lattice $U(2)^r$ ($\varepsilon = 0$) or $U(2)^{r-1} \oplus D_4$ ($\varepsilon = 1$); the Milgram phase is $4 \cdot \text{Arf} \bmod 8$; the Weil S -matrix prefactor is $(-1)^{\text{Arf}}$; and the Frobenius–Schur

indicator of the Heisenberg representation of E_Q is $(-1)^{\text{Arf}}$. The metaplectic relation $(ST)^3 = S^2$ holds iff the diagonal T uses a quadratic refinement of the correct Arf class – it selects the Arf *class*, not the member. So the integral bridge carries exactly one game-relevant bit; it cannot, by itself, distinguish the Gold form from any other refinement in its class.

7.5 A Tier-2 naturality screen: E -equivariance

Proposition 1 kills rules that are blind to everything but the symplectic structure (Tier 1), while per-position evaluator circuits for Q_a realize the quadric tautologically (Tier 3). The extraspecial dictionary suggests a criterion strictly between the two: the rule should see the extension E_Q – which carries the diagonal data q_i structurally, as squares – but only up to its automorphisms, so that no basis, field structure, or evaluation circuit can be smuggled in.

Definition 1 (uniform rules; the three tiers). Let Q be nondegenerate on $V = \mathbb{F}_2^{2r}$ with polar form B , and $E = E_Q$, $\pi: E \rightarrow V$, $\Sigma = \{x \in V : Q(x) = 0\}$.

- (a) A *uniform rule on a finite set* X is a single move relation $M \subseteq X \times X$, fixed once for all positions; for normal play we require the move digraph acyclic, and outcomes are computed by the usual retrograde recursion. (For loopy or misère readings, replace the outcome recursion; the equivariance constraint below applies verbatim, since outcome classes are invariants of digraph automorphisms.)
- (b) A uniform rule on E is *E -equivariant* if every $\alpha \in \text{Aut}(E)$ is an automorphism of the move digraph. Its *shadow* is $\pi(P) \subseteq V$, where P is its P -set; it *realizes* $T \subseteq V$ if $\pi(P) = T$.
- (c) *Tier 1* ($\text{Sp}(B)$ -blind): a uniform rule on V invariant under $\text{Sp}(B)$, as in Proposition 1. *Tier 3* (Q_a -evaluation): a family of games $\{\Gamma_x\}_{x \in V}$ whose structure varies with the designated input x (e.g. the trace-circuit evaluator); this is not a uniform rule. *Tier 2 screen*: a route to the Gold quadric is Tier-2 natural *only if* its positions and moves lift to an E -equivariant uniform rule on E_{Q_a} – with Q_a restricted to its nonsingular core when the form is degenerate, matching the classifier’s radical discipline – that realizes $\{Q_a = 0\}$.

Proposition 3 (the screen sits strictly between the solved tiers). Let Q be nondegenerate on $V = \mathbb{F}_2^{2r}$, $r \geq 2$, and $E = E_Q$.

- (i) The image of $\text{Aut}(E) \rightarrow \text{GL}(V)$ is exactly the orthogonal group $\text{O}(Q)$, with kernel the central twists $g \mapsto g z^{\ell(\pi g)}$, $\ell \in V^*$, which equal $\text{Inn}(E) \cong V$ because B is nondegenerate; thus $\text{Out}(E) \cong \text{O}(Q)$ (cf. [15]). The $\text{Aut}(E)$ -orbits on E are

$$\{1\}, \quad \{z\}, \quad \pi^{-1}(\Sigma \setminus \{0\}), \quad \pi^{-1}(V \setminus \Sigma).$$

- (ii) Hence the P -set of any E -equivariant uniform rule is a union of these four orbits, and its shadow is one of the eight unions of $\{0\}$, $\Sigma \setminus \{0\}$, $V \setminus \Sigma$. The quadric Σ is among them: the Tier-1 exclusion does not apply.
- (iii) (Tier 1 $\subsetneq$ Tier 2.) Every $\text{Sp}(B)$ -invariant uniform rule on V pulls back along π to an E -equivariant rule with P -set π^{-1} of the original, whose shadow is therefore one of $\emptyset$, $\{0\}$, $V \setminus \{0\}$, V – never the quadric (Proposition 1 in this language). The containment is strict: the squaring rule

$$g \rightarrow h \text{ legal} \iff g^2 = z \text{ and } h^2 = 1$$

(“move from any order-4 position to any position of order at most 2”) is E -equivariant and acyclic, and its P -set is the involution locus $\{g : g^2 = 1\} = \pi^{-1}(\Sigma)$, with shadow exactly Σ .

(iv) (*Tier 2* $\subsetneq$ *Tier 3*.) Up to the isometry induced by any group isomorphism, the family of E -equivariant rules and their shadows depends only on $(r, \text{Arf } Q)$ (Lemma 2(iii)). In particular the screen cannot separate Gold forms of equal core rank and Arf invariant, and an E -equivariant rule has no access to m , the exponent a , the field multiplication, the coordinate frame $q_i = Q_a(e_i)$, or any evaluation circuit. Conversely, arbitrary subsets of V – all realizable by ad hoc acyclic lookup games and by Tier-3 evaluators (Section 6.1) – are not shadows of E -equivariant rules once they leave the eight-set list of (ii).

Proof. (i) Automorphisms preserve $Z = Z(E)$ and fix z ($\text{Aut}(\mathbb{Z}/2) = 1$), so the induced map preserves the squaring form: $Q(\bar{\alpha}x)$ is read from $\alpha(\hat{x})^2 = \alpha(\hat{x}^2) = z^{Q(x)}$; the image lies in $O(Q)$. Surjectivity: an isometry g transports the standard cocycle as in the proof of Lemma 2(iii), producing an automorphism of E_Q inducing g . The kernel consists of maps $g \mapsto g z^{\ell(\pi g)}$ with ℓ linear (multiplicativity forces additivity of ℓ); inner automorphisms are exactly the twists $\ell = B(\bar{g}, \cdot)$, and nondegeneracy makes $\bar{g} \mapsto B(\bar{g}, \cdot)$ onto V^* . Orbits: $O(Q)$ is transitive on $\Sigma \setminus \{0\}$ and on $V \setminus \Sigma$ by Witt's extension theorem in its characteristic-2 quadratic-space form [14] (both sets are nonempty for $r \geq 2$); the two lifts of any $x \neq 0$ are conjugate under conjugation by $\tilde{g}$ with $B(\bar{g}, x) = 1$; and $1, z$ are fixed by everything.

(ii) Digraph automorphisms preserve the retrograde outcome recursion, so P is $\text{Aut}(E)$ -invariant; apply (i) and project.

(iii) The pullback $M = \{(g, h) : (\pi g, \pi h) \in R\}$ of an $\text{Sp}(B)$ -invariant rule R is preserved by $\text{Aut}(E)$, since the induced action on V lies in $O(Q) \subseteq \text{Sp}(B)$; it is acyclic when R is, and an easy induction along the acyclic rank gives $P(M) = \pi^{-1}(P(R))$. By the transitivity of $\text{Sp}(B)$ on $V \setminus \{0\}$ for $r \geq 2$, $P(R)$ is a union of $\{0\}$ and $V \setminus \{0\}$. For the squaring rule: automorphisms preserve element orders, so the rule is E -equivariant; positions with $g^2 = 1$ are terminal, hence P ; positions with $g^2 = z$ have a move (to 1), hence N . Its shadow Σ satisfies $1 < |\Sigma| = 2^{2r-1} + \varepsilon 2^{r-1} < 2^{2r} - 1$ for $r \geq 2$, so it is none of the four Tier-1 shadows.

(iv) If $\psi: E_Q \rightarrow E_{Q'}$ is an isomorphism (it exists iff ranks and Arf agree, Lemma 2(iii)), then $M \mapsto (\psi \times \psi)(M)$ is a bijection between E -equivariant rules carrying P -sets to P -sets and shadows to shadows through the induced isometry ψ ; in particular it carries rules realizing $\{Q = 0\}$ to rules realizing $\{Q' = 0\}$. The shadow constraint of (ii) excludes all but eight subsets, while Tier-3 constructions realize any subset; the containment is proper. $\square$

Remark 8 (what the screen does and does not settle). Equivariance is a screen, not a construction. Over the *abstract* group E the screen is satisfiable – Proposition 3(iii) exhibits the squaring rule, which is nothing but the extraspecial squaring map of Lemma 1 read as a one-move relation. That rule consults only the multiplication of E , so it is a Q -evaluator exactly insofar as E itself is fed in by hand. Historically the reframing relocated the problem: instead of *find a play rule that computes Q_a* , it asked *exhibit a game-native model of the extension E_{Q_a}* – positions built from game values, multiplication from game constructions – after which the rule layer is canonical. Lemma 3 is the sharp constraint on that search: no commutative value-world, in particular no misère-quotient kernel (Corollary 1), can host E once $B \neq 0$. The noncommutativity must enter from a structurally available asymmetry, and the one that normal, misère, and partizan play all possess – and that the symmetric polar form B discards – is the first-/second-player asymmetry of the move relation. Claim levels: the proposition is standard mathematics; treating E -equivariance as *the* naturality criterion is interpretation. The weighted-source rule resolves the P -set problem covariantly without first constructing the entire abstract extension as a game object; such a full model remains a stricter optional generalization. (For $r = 1$ the screen degenerates: on the anisotropic plane $\Sigma = \{0\}$ and $O(Q) = \text{Sp}(B)$, so the statement is vacuous there; the Gold targets of interest have $r \geq 2$ cores.)

Theorem E below sharpens the screen further: read as *invariance* it admits no live middle at all, so the E -structure must enter *covariantly*, in the dynamics.

Open question. *Does any game-native construction realize the extraspecial extension E_{Q_a} – equivalently, produce the diagonal data $q_i = Q_a(e_i)$ as squares in a noncommutative game-built structure – without an evaluation circuit for Q_a ? By Lemma 3 the source cannot be any commutative game-value monoid.*

8 No-go theorems for the middle tier

This section records the no-go results of the structured attack (2026-06-10) on the middle tier. Architectures covered: commutative provenance, affine and orthogonal symmetry, bounded B -oracle access, B -local flip rules, and E -translation dynamics. Verification scopes are stated inline; the supporting probes live in `experiments/gold/`. (Theorem A, coin-turning linearity, opened the series in Section 5.)

Theorem B (commutative squaring collapse; standard math). *In any commutative monoid, squaring is an endomorphism, so any quadratic datum read through homomorphic coordinates has polarization identically zero. In particular the extraspecial squaring map Q (for $B \neq 0$) cannot live in, map into, or pull back through any commutative game monoid. This covers misère quotients, turn-parity counters, and all coin-turning configuration sums.*

Proof. Immediate from Lemma 3: commutativity gives $(mn)^2 = m^2n^2$, so the squaring form is additive and $B \equiv 0$. $\square$

Theorem C (group misère quotients are tiny; conditional). *Assume [12, Thm. 6.4] with its regularity hypothesis. If the misère indistinguishability quotient $\mathcal{Q}(\mathcal{A})$ of a set of impartial games is a group, then $|\mathcal{Q}(\mathcal{A})| \leq 2$.*

Proof sketch. Doubling lemma: in a group quotient every element is cancellable, and any position of Grundy value 2 doubles to a misère- P position, contradicting Grundy-distinctness from 0; the quoted structure theorem then pins the quotient to at most $\{1, a\}$. $\square$

Remark 9 (scope and a probe caveat). Theorem C is conditional on the Plambeck–Siegel structure theorem as cited; the regularity hypothesis is quoted from the repository’s `experiments/misere_kernel.py` docstring, and the JCTA 2008 paper has *not* been independently re-verified within this program (closing that gate is a listed next move). Two consequences for the probe map: the $(\mathbb{Z}/2)^k$ ($k \geq 2$) target of `examples/octal_hunt.rs` is empty a priori – its empirical negatives (quotient orders 2, 6, 10, 12, 14) are a theorem, not a sampling accident – and any future “hit” from that sweep would be a labeling artifact, since the helper `p_set_as_f2` checks only that the subset labeling is a bijection, not that it is a monoid homomorphism. The hunt’s target should be reframed accordingly.

For the remaining results, fix a nondegenerate quadratic form Q on $V = \mathbb{F}_2^{2r}$, $r \geq 2$, with polar form B and zero set $\Sigma = \{Q = 0\}$, and write $\text{AO}(Q) = \{x \mapsto gx + c : g \in \text{Sp}(B), Q(gx + c) = Q(x) \forall x\}$ for the affine stabilizer appearing below.

Theorem D (affine symmetry budget). *$\text{Stab}_{\text{AGL}(V)}(\Sigma) = \text{AO}(Q)$; it contains no nontrivial pure translation (for a nonsingular core, $c \in \text{rad}(B) = 0$), and its pure-linear part is exactly $\text{O}(Q)$. Consequently every uniform rule whose move digraph admits even one affine automorphism outside $\text{AO}(Q)$ has P -set $\neq \Sigma$.*

Verification. Exhaustive over all 322,560 affine maps of $\mathbb{F}_2^4$ at $m = 4$, both Arf classes (`experiments/gold/skeptic_nogo_check.py`); the linear-part identification is standard math (Witt extension).

Theorem E (the symmetry dial has no middle). $O(Q)$ is maximal in $\mathrm{Sp}(B)$, and the $O(Q)$ -orbitals on $V \times V$ are exactly the Gram classes $(Q(u), Q(w), B(u, w))$ together with the degeneracy flags (Witt extension; see [7, §8]). Hence a rule invariant under any group strictly larger than $O(Q)$ is dead by Theorem D (its symmetry reaches outside $\mathrm{AO}(Q)$), while a rule fully $O(Q)$ - or $\mathrm{Aut}(E)$ -equivariant factors extensionally through pointwise Q - and B -evaluation. There is no invariance class strictly between dead (too coarse) and Gram-factoring (an evaluator): E -naturality must be covariance, not invariance.

Verification. Maximality verified at $m = 4$: every one of the 648 (resp. 600) elements of $\mathrm{Sp}(4, 2)$ outside $O^+(4, 2)$ (resp. $O^-(4, 2)$) generates $\mathrm{Sp}(4, 2)$ together with the orthogonal group; orbitals enumerated exhaustively at $m = 4$ (`experiments/gold/ao_orbitals.py`).

Theorem F (B -oracle lower bound). Access model: legality of a move is decided from XOR combinations of the two positions and t fixed constant vectors, with oracle access to B only. If $t \leq 2r - 3$, then for every quadratic refinement Q' of B simultaneously there is a nonzero Q' -singular vector in the complement of the constant span; the transvection it induces is a digraph automorphism lying outside $O(Q')$, and Theorem D kills the P -set. The bound is tight: at $t = 2r - 2$ with anisotropic complement the argument fails. Hence the Gold diagonal framing $q_i = Q_a(e_i)$, which supplies $2r$ frame constants, is within two dimensions of information-theoretically forced.

Verification. Transvection step and the exact escape boundary $t = 2r - 2$ machine-checked at $m = 4$; end-to-end spot check on 50 random $t = 1$ B -oracle rules, retrograde-solved (`experiments/gold/skeptic_nogo_check.py`, `nogo_verify.py`).

Theorem G (B -local flip rules). Any rule of the form “flip d at v iff $f(d, B(v, d))$ ” is automatically undirected, because B is alternating: $f(d, B(v + d, d)) = f(d, B(v, d) + B(d, d)) = f(d, B(v, d))$. For undirected loopy games the outcome partition collapses: Win = $\emptyset$, Loss = the isolated vertices = an affine flat, and Loss $\cup$ Draw $\neq \Sigma$ for $r \geq 2$ (the count $|\Sigma| = 2^{r-1}(2^r + \varepsilon)$ has an odd factor > 1). This covers the Loss, Draw, and Loss $\cup$ Draw targets at once. The symmetric- B rule of `examples/loopy_quadric.rs` is the special case $f(d, \beta) = \beta$; its $(4, 1)$ “hit” is the radical coincidence of Section 6.1.

Verification. Undirectedness is the two-line identity above (standard math); the loopy outcome collapse including the Draw-set gap is machine-checked at $m = 4$ (`experiments/gold/skeptic_nogo_check.py`).

Theorem H (E -translation no-go). For $r \geq 2$, no left- E -equivariant rule on $E = E_Q$ with moves $g \rightarrow gn$, $n \in N$ for a fixed nonempty $N \subseteq E$, has P -set equal to the involution locus $I = \{g : g^2 = 1\} = \pi^{-1}(\Sigma)$.

Proof. $I \cdot I = E$: for every $v \neq 0$, the count $\#\{x : Q(x) = 0 = Q(x + v)\} = 2^{2r-2} + (-1)^{\mathrm{Arf} Q} 2^{r-1}$ is positive for $r \geq 2$ (character sum), so every $n \in N$ factors as $n = gh$ with $g, h \in I$. Then $g \in I$, and $g \cdot n = g(gh) = g^2h = h \in I$: the move $g \rightarrow gn$ is an edge from I to I . A normal-play P -set admits no P -to- P edge, so $P \neq I$. $\square$

Verification. Cross-checked on the bent $(8, 1, \lambda = 2)$ extraspecial group (`experiments/gold/extraspecial_core.py`).

8.1 Normal-play rigidity and the synthesis

The strongest constraint is not any single architecture kill but a rigidity statement inside a declared access class. Call a rule *in-class* if the quadratic refinement is quarantined behind a weight-bounded oracle (the (w_0, c) metering of Definition 2 below) and the rule is *refinement-uniform*: it realizes the target for every refinement q of B in the declared family, not just the Gold one.

Theorem 2 (normal-play rigidity, in-class). *Under the bounded-framing oracle model with refinement uniformity, every legal edge between positions of Hamming weight $> cw_0$ must flip Q – in normal, misère, and loopy-Loss semantics alike. Consequently every in-class realizer of Σ in those semantics is a Q -alternating ender: play telescopes Q , and the outcome is the parity of $Q(x)$.*

Proof sketch. The case $Q(v) = Q(w) = 0$ of a non-flipping edge is forbidden directly by the outcome structure in all three readings. The case $Q(v) = Q(w) = 1$ is reduced to it by an adversary substitution: replace q by $q' = q + \ell$ with ℓ a linear form annihilating every queried functional but not the dense indicators; refinement uniformity transports the rule to q' , where the edge becomes a forbidden 0-to-0 edge. $\square$

Theorem 3 (no live middle, relative to the quarantine). *Let Q have nonsingular core of rank $2r \geq 4$. No fixed uniform single-board rule satisfying*

- (S) *no affine digraph automorphism outside $\text{AO}(Q)$,*
- (A) *legality from B -oracle queries at $t \leq 2r - 3$ frame constants,*
- (C) *no commutative squaring route for the provenance of Q ,*
- (E) *no left- E -equivariant structure on the extraspecial group,*
- (R) *bounded-framing oracle access with refinement uniformity,*

has $P/\text{Loss}/\text{Draw}$ target equal to Σ with any bulk strategic content: under (R), every legal edge between dense positions flips Q , all strategic content is confined to a Hamming ball of radius cw_0 , and no bulk-decision-live realizer exists.

Remark 10 (honest scope: five historical escape hatches). Theorem 3 is relative to its hypotheses, and each hypothesis exposed an experimental window during the no-go phase:

1. *Loopy-Draw semantics*: the substitution contradiction of Theorem 2 does not fire (Draw-to-Draw edges are legal).
2. *$t \geq 2r - 2$ with anisotropic complement*: escapes Theorem F at the symmetry level.
3. *Frobenius-aware access*: the Galois group lies inside $\text{O}(Q_a)$, so all symmetry methods are silent; Theorem F’s model explicitly excludes Frobenius-twisted queries.
4. *Non-quarantined rules using the game-native $\wp(w)$ diagonal source* (Proposition 2): once q is game-built, refinement uniformity holds trivially and class (R) says nothing.
5. *Rank $r = 1$ and radical-anisotropic degenerate layers*: the character-sum count in Theorem H vanishes, and small extraspecial groups can be realized.

The quarantine hypothesis (R) is a choice with a justification: without quarantining the form-identifying datum, the descent evaluator is decision-live everywhere and no purely complexity-based no-go is possible. Whether game-built diagonal sources such as $\wp(w)$ count as “probing” or “feeding in” the form was the central definitional question of that program. Section 9 replaces the vocabulary quarantine by the invariant transcript-span boundary, and Section 11 attains it with an explicit local rule.

9 Local realization and the observation boundary

Two natural first attempts at “naturalness” fail for structural reasons. *Equivariance fails*: by Theorem E there is no invariance class strictly between symmetry groups that kill the P -set and symmetry groups that force extensional factoring through Q - and B -evaluation. *Vocabulary quarantine fails*: the Gold diagonal $q_i^{(m,1)} = \text{Tr}(e_i^3)$ is expressible as $\text{Tr}(\wp(w)e_i)$ using only XOR, nim-product, and trace (Proposition 2) – the licensed operation chain itself – so no principled line separates “computing q ” from “computing Q ” on vocabulary alone. The criterion below replaces both with access metering and a sharp transcript-observation boundary.

Definition 2 (uniform local realization). Fix the coin frame $\{e_i\}$ (the Grundy basis $g(n) = 2^n$, game-natural). *Public data*: the frame, $\oplus$, nim-product, Frobenius, and the full alternating form B (game-built via Turning Corners, Frobenius, and trace). *Quarantined data*: the quadratic refinement, presented as a weight-bounded oracle $O_q(z) = Q_q(z)$ for $\text{wt}(z) \leq w_0$. A *uniform local rule* satisfies:

- F1** (*q-blind statics*) position set, loading map, and terminal set are computed without oracle access;
- F2** (*metered access*) each move predicate and state transition together use at most c adaptive oracle queries, each at a point of Hamming weight at most w_0 , with (w_0, c) constants independent of m ;
- F3** (*declared semantics*) one canonical outcome labeling (normal, misère, loopy, or q -blind terminal payoff).

It is an *exact local realizer* if additionally:

- N1** (*torsor uniformity*) for every m , every B in the declared family, and every refinement q of B : $\text{outcome}(\iota(x)) = \text{target}$ iff $Q_q(x) = 0$;
- N2** (*locality budget*) the (w_0, c) metering of F2;

A rule with a reachable position having both a winning and a losing move is called *decision-nondegenerate*. These are objective properties of a declared rule and oracle interface. They do not, by themselves, turn the informal adjective “natural” into a semantic invariant.

Theorem 4 (Tier-3 exclusion under N1+N2). *For any position x with $\text{wt}(x) > cw_0$, the at most c queried functionals span only weight- $\leq cw_0$ directions, so there exist refinements q, q' agreeing with every oracle answer but differing on $Q(x)$. Hence no in-class rule’s move predicate factors through $Q(x)$ at dense positions – by theorem, not stipulation.*

Definition 3 (bounded outcome certificate). Write $\text{N2cert}(C, w)$ when, for every refinement q and input x , there is a finite observation set $S(q, x)$ with $|S(q, x)| \leq C$ and $\text{wt}(z) \leq w$ for $z \in S(q, x)$ such that every same-polar refinement q' agreeing with q on $S(q, x)$ has the same rooted outcome. The set may be chosen adaptively from q ; what is bounded is the complete certificate for the outcome, not the number of queries made by one transition.

Theorem 5 (sharp observation lower bound). *Let the refinement family at fixed B be closed under addition of every linear functional. If a transcript-stable rooted result is exact at x , then*

$$x \in \text{span } S(q, x).$$

In the original frame, if every observed vector has weight at most w , then

$$\text{wt}(x) \leq |S(q, x)|w.$$

Consequently no constants C, w give an exact torsor-uniform family satisfying $N2cert(C, w)$ in unbounded dimension.

Proof. If $x \notin \text{span } S$, finite-dimensional separation gives a linear functional ℓ vanishing on S but not at x . The refinements q and $q + \ell$ therefore have identical observation transcripts and rooted outcomes, while their target values at x differ, contradicting exactness. For the coordinate bound, the union of the supports of $|S|$ weight- w vectors has size at most $|S|w$ and must cover the support of x . Both forms are kernel-checked in `formal/Ogdoad/GoldNoEvaluator.lean`. $\square$

Theorem 6 (fork padding obstruction). *The properties “there exists a refinement-sensitive winning fork at a reachable position” and the stronger variant requiring that fork after every original completed (hence every optimal completed) play do not certify that a finite normal-play realizer is non-evaluative. Every finite normal-play tree can be padded, without changing its root outcome, so that every completed play enters an always- N two-action fork whose unique winning action swaps with one chosen refinement bit.*

Proof. Replace every terminal P -node by a one-option wrapper whose child is the always- N swapping fork. The wrapper is still P , so induction over the finite tree preserves every ancestor outcome. The new fork is forced after every original completed play. The construction and induction are kernel-checked as `win_padTerminals` in `formal/Ogdoad/GoldForkPadding.lean`. $\square$

Remark 11 (Why the old N3 screens are retired). The old mixed-successor N3 is defeated by dominated escape edges, while after dominance pruning every retained successor at a position has one outcome class. Counterfactual winning-action variants remain vulnerable to Theorem 6, even if their witness is required on every optimal play. Thus the paper uses the representation-invariant certificate lower bound of Theorem 5 for the negative boundary and states the positive rule’s syntax and access discipline directly. Strategic liveness remains useful descriptive evidence, not an axiom defining naturality.

Calibration (implemented and tested at the stated instances). Under F1–F3 + N1–N2, the T2 spin-flip rule and the pending-marker clock are exact serial evaluators; ECHO-ko passes the weaker old N3 (at least 160 mistake-states at each verified exact $m = 4$ instance); an order-4-alphabet loopy candidate from the run fails N2 (its alphabet is a 2^m -point Q -evaluation); an unwinding-game candidate from the run fails N1 (isotropic frames break exactness); and the repository’s Tier-3 circuit and `interactive_kernel` Rule 2 fail N2 (dense-point form access). The candidates named here live in `experiments/gold/`. None of these finite calibrations substitutes for the certificate theorem.

10 Constructions and near-misses

10.1 T2: the width-2 spin-flip rule (exact, but a clock)

Proposition 4 (T2 realizes every char-2 quadric; implemented and tested). *On positions $x \in \mathbb{F}_2^n$, let the rule be: turn over a set d of coins with $\text{wt}(d) \in \{1, 2\}$, leading coin heads-to-tails, legal iff $B(x, d) \oplus Q(d) = 1$. Then for every characteristic-2 quadratic form Q , the normal-play P -set is exactly $\{Q = 0\}$.*

Proof sketch (blocking lemma). Every legal move flips Q (by the polarization identity, since $Q(x + d) = Q(x) + Q(d) + B(x, d)$ and legality forces $Q(d) + B(x, d) = 1$), so it suffices that every position with $Q(v) = 1$ has a legal descent. If all descents from v fail, then $B(v, e_i) = q_i$ for all

$i \in \text{supp}(v)$ and $B \equiv 0$ on $\text{supp}(v) \times \text{supp}(v)$; expanding $Q(v)$ in the basis then forces $Q(v) = 0$, a contradiction. $\square$

Verification. Checked at $(8, 1)$, $(16, 1)$, the bent $(8, 1, \lambda = 2)$, and 20 random refinements of each B (the “refinement uniformity certificate”; `experiments/gold/witness_test.py` and companions). For $a = 1$ the rule constants q_i are game-native via Proposition 2; for every a they are constructively game-built by Theorem 1. T2 nevertheless remains a forced clock.

T2 is, however, a forced clock: every legal move flips Q , play telescopes $Q(x)$ as a parity count, every position’s options share one outcome class, and no position has a losing move. By Theorem 2 this is unavoidable for *all* in-class normal-play realizers, not a defect of T2 specifically. T2 and the pending-marker clock pin the boundary: within every tested bounded-access quarantine, exact realizers exist but are forced clocks.

10.2 echo-ko: charge counting on the extraspecial cocycle

The one known exact realizer that is *not* a clock works one level up, on the extraspecial cocycle itself.

Rule (echo-ko). Positions are indexed by $x \in \mathbb{F}_{2^m}$; the coins of x must each be touched twice; players alternate single touches; each touch of e_i updates a charge $\sigma \oplus = c(\text{open-set}, e_i)$, where c is the triangular cocycle of the extraspecial extension ($c(v, v) = Q_a(v)$, $c(u, v) \oplus c(v, u) = B(u, v)$, cf. Lemma 1(ii)); a one-move ko forbids immediately re-touching, a player with no legal touch passes (the pass clears the ko), and play ends when every coin is closed. The two players are the two central characters: the complete play word lies over $0 \in V$, hence equals 1 or z in E_Q ; one player plays for terminal charge $\sigma = 1$ (word = z), the other for $\sigma = 0$, and the *value* of the position is the forced terminal charge.

The rule is center-reading (not center-blind), noncommutative via move order (not turn count), and – at every verified exact instance – decision-nondegenerate. For $a = 1$ its constants are game-built via Proposition 2, with no per-instance Q -evaluation.

Provenance and results (implemented and tested). A first-round solver had a bug (the memoization key omitted the accumulated charge parity); the corrected solver was validated against explicit tree enumeration. Corrected results (round-1 apparatus: `experiments/gold/echo_charge_probe.py` and companions — the rescued snapshot of that file predates the memo-key fix, so cite `experiments/echo_solver.py` stage `ko2` below, which reproduces this table exactly):

instance	core	agreement	decision-live
$m = 4$, bent $\lambda \in \{2, 12\}$	rank 4	16/16 exact	yes (≥ 160 mistake-states)
$(8, 2)$, $\lambda = 1$	rank 4	255/256 (miss $x = 224$)	yes
$(8, 1)$, $\lambda = 1$	rank 6	228/256	yes
$(8, 1)$, bent $\lambda = 2$	rank 8	212/256	yes

Agreement grades by polar rank. The table reports the $\sigma=1$ -stance face (first player plays for z) of a stance-asymmetric rule: with the first player playing for $\sigma = 0$ the same instances degrade (15/16 at the bent $m = 4$ instances, 250/256 at $(8, 2, 1)$). An independent σ -explicit full-state direct solver, validated against tree enumeration (`experiments/echo_solver.py`, stage `ko2`; 2026-06-10), reproduces the table exactly, including the unique $(8, 2)$ miss $x = 224$ – so the corrected round-1

numbers now have an independent re-derivation. The FIFO+dummy variant below is, by contrast, exact at *both* stances. A bounded-memory blocker conjecture (for fixed ko-window w , adversarial unlinking wins on supports $k \geq w + 2$ at rank ≥ 6) was formulated on the *invalid* round-1 data and has not been re-examined against the corrected solver; it is recorded as a hypothesis to re-test, not a finding.

10.3 echo-fifo+dummy: the verified exact $m = 8$ realizer

A second-round variant imposes a FIFO close-discipline on top of the rule above – only the longest-open coin may be closed, while the one-move ko and the forced pass are *retained* – and adjoins one neutral tempo coin ($q = 0$, zero polar row) to every board, fully queued and ko-able. The readout is σ -valued as in ECHO-ko, at both stances $t \in \{0, 1\}$; “exact” means the forced terminal charge equals $Q(x)$ for every position and both stances (765 scaled forms $\times$ 256 positions $\times$ 2 stances = 391,680 checks). The original run established this via a decomposition-plus-isomorphism-caching solver validated only at $m = 4$, and earlier drafts recorded the claim as unverified.

Adversarial review (2026-06-10; implemented and tested). The pre-registered experiment of Section 10.4 was executed with a fresh direct stateful solver (`experiments/echo_solver.py`): no decomposition, no isomorphism caching, full state – untouched set, open queue in order, ko memory, mover, and the accumulated charge σ – in the memoization key. Validation: explicit no-memo tree enumeration ($m = 4$ exhaustive, $m = 8$ small supports); the original direct $m \leq 4$ solver executed verbatim as a cross-oracle (1,920 solves agree); nim arithmetic against the independent Turning-Corners mex recurrence; and a second reviewer model re-ran every stage and re-checked the nim product against the original probe’s implementation on all 65,536 pairs. Results:

- $m = 4$ family: exact at all 15 scaled forms, both stances, with the dummy (30/30; only 15/30 without it – the tempo coin is load-bearing).
- $m = 8$ benchmarks (8, 2, 1) rank 4, (8, 1, 1) rank 6, (8, 1, 2) and (8, 1, 3) rank 8: 256/256 at both stances.
- Rank-stratified sample (21 (a, λ) pairs across ranks $\{4, 6, 8\}$): exact, both stances.
- The full claimed sweep, re-derived end to end: **391,680/391,680**, zero misses.
- Decision-liveness: 1.46–4.37 million decision states per benchmark instance (128–210 of 256 positions decision-live) – not a clock.
- Torsor sweep: 20 refinements of each benchmark polar form (Gold, zero, all-ones, seeded-random diagonals), exact throughout. Honest note: each q_i fires exactly once, at the close of coin i , so the diagonal contribution to σ is play-invariant; both-stance exactness at one refinement of B therefore *implies* it for all refinements, and the sweep is a consistency check of that argument, not independent evidence.

Historically, this established the exactness of the first σ -valued realizer but left two boundary questions. Its charge had not yet been recast as a normal-play P -set, and its proof was bounded to the computed window. The later phase-aware claim compiler closes the first gap, while the Witt-matching theorem and weighted-source replacement close the second without a tempo coin. The bounded-window architecture remains of historical interest, but it is no longer load-bearing for existence.

10.4 The decisive experiment

The pre-registered experiment for the primary candidate: a corrected-solver sweep over the finite ECHO rule family at $m = 8$, validated by a direct stateful solver.

- *Benchmarks:* $(8, 1, \lambda = 1)$ rank 6; bent $(8, 1, \lambda = 2)$ rank 8; $(8, 2, \lambda = 1)$ rank 4; the full $m = 4$ family as regression.
- *Axes:* ko-memory window $w \in \{1, 2, 3\}$; pass semantics (clears-ko / forbidden / loses); single-coin plus pair touches (the tartan-companion axis); both orientations.
- *Success criterion:* zero misses across all tested (m, a, λ) triples, decision-nondegenerate throughout, and surviving a torsor sweep (≥ 20 stratified refinements of each B , confirming refinement uniformity rather than single-refinement luck).

A CONFIRM outcome would have been the first genuine Tier-2 witness, pending the then-open outcome recasting and even- a diagonal lemma (both closed later in this note and its diagonal-source companion). A KILL outcome – rank-graded decay persisting across every family member – would put the bounded-memory blocker conjecture on valid data and close the line for bounded-memory architectures.

Executed 2026-06-10 (Section 10.3): outcome CONFIRM on every leg – zero misses, decision-nondegenerate throughout, torsor-uniform. The remaining family axes (ko-memory window $w \in \{1, 2, 3\}$, pass semantics, pair touches) are no longer decisive for existence; they map the boundary of the mechanism – which disciplines besides FIFO+dummy are exact, and why – and feed the reconciliation item of Section 12.

10.5 The linking reduction and the general- m theorem

The mechanism question of the verified realizer reduces, by three short play identities, to one combinatorial statement about graphs (`experiments/linking_game.py`; second-pass run 2026-06-10). Fix a board: coins = vertices of a graph G (on a Gold board, G is the polar graph B restricted to $\text{supp}(x)$, plus the dummy as an isolated vertex; the diagonal q is play-invariant and splits off, as in the torsor note of Section 10.3).

Reduction (standard math; each step is a whole-play identity, machine-validated on random plays).

1. *No nesting.* FIFO forces coins to close in opening order, so no open-window nests inside another; an edge of G is linked iff its two windows *overlap*, and σ equals the parity of $E(G)$ -edges covered by the play’s interval-overlap graph.
2. *Odd-close accounting.* Charging an edge when its second endpoint opens (same play total), set $\text{und}(s)$ = the number of edges with an untouched endpoint. Then $D = \sigma \oplus \text{und}$ is invariant under opens and passes, and flips exactly when the queue front f closes with $\deg_U(f)$ odd (U = untouched set). The whole game is therefore the *odd-close parity game*: D starts at $|E| \bmod 2$, only odd-degree front closes flip it, and $\sigma_{\text{final}} = D$.
3. *Ko localization.* Opens of untouched coins are never ko-blocked; a close is ko-blocked only when the front was opened onto an empty queue on the immediately preceding move; forced passes occur only once $U = \emptyset$, after which no flip is possible. Passes are irrelevant to the flip fight.
4. *Block turns.* A maximal nonempty-queue block containing b coins has exactly b opens and b closes, hence $2b$ touches. The initial mover therefore starts every block. Before the single possible terminal pass, queue length is even on that player’s turns and odd on the other player’s turns.

5. *Live-degree accounting.* Let Q be the queued set, U the untouched set, $L = Q \cup U$ the still-unclosed set, and $P = e(Q, U) \bmod 2$. Opening x changes P by $\deg_L(x)$; closing any queued f changes P by $\deg_U(f)$, exactly the flip bit. Since P starts and ends at zero,

$$\# \text{flips} \equiv \sum_{x \text{ opened}} \deg_{L_x}(x) \pmod{2}.$$

Equivalently, for $S = (\text{flips so far}) + P$, closes are free and an open scores the live-degree parity of its vertex. The identity is FIFO-blind—it holds if any queued coin may close—so a proof must put FIFO back through the fact that the close target is the front.

The linking theorem (target). *If the board contains an isolated coin, the flip count is forced even – from both seats, for every graph.* Equivalently σ is forced to $|E| \bmod 2$, which on a Gold board is $Q(x)$: m -uniform exactness of ECHO-FIFO+dummy.

Verified (implemented and tested). The statement holds for *every* isomorphism class on $k \leq 8$ real coins plus dummy, both seats (12,346 classes at $k = 8$, generated by nauty **geng**) – strictly stronger ground than the Gold-arising boards of the $m = 8$ sweep. The reduced solver now stores ko as the one bit it is rather than the last coin’s identity; exhaustive cross-checks against the original σ -explicit solver pin that reduction. Without the isolated coin the failures (“Bad graphs”, census 1/4/34 at $n = 3/5/7$, none at even n) are always *mover-controlled* – the first player forces whichever parity they want, never the anti-parity – and none contains an isolated vertex; 33/34 at $n = 7$ have a dominating vertex.

Device lemma (standard math). With the queue empty, if some $v \in U$ dominates $R = U \setminus \{v\}$ with $|R|$ even and nonempty, the mover forces a flip in two plies: open v (ko-protected, $\deg_U(v)$ even), every reply must open some $y \in R \subseteq N(v)$ making v odd, close v . Conversely, a single odd front at the responder’s turn is always re-evenable (odd degree ≥ 1 supplies a never-ko-blocked neighbor open), and when v does not dominate R a non-neighbor open preserves evenness. An isolated coin defeats this empty-queue device at every root (nothing dominates a set containing it), matching the Bad census; for $|R|$ odd, domination makes $\deg$ odd and the neighbor reply works, explaining the bonus rigidity of all even- n boards.

The device is not the unique local squeeze. On the path $z - f - y - h$, take queue (f, h) and $U = \{y, z\}$ on the defender’s turn. The front has even degree two; opening either untouched vertex makes it odd, while closing it exposes h with odd degree one. Thus every move offers the attacker an odd close despite the nonempty queue. An untouched isolated coin kills this squeeze too, but a queued or spent dummy does not.

The finite strategy boundary. A two-mode menu strategy tracking the flip debt g – PREVENTION ($g = 0$: re-even odd fronts; non-toggling opens, e.g. the dummy; safe closes; in the trap $U \subseteq N(\text{front})$, poison or close) and DEBT ($g = 1$: counter-close odd fronts – the attacker’s flip-close typically exposes the repair; else toggle or advance) – does beat an optimal unrestricted attacker on every class $k \leq 7$, both seats, with no move outside the menus, but it does *not* extend as written. At $k = 8$ graph6 class **GCRU]w** (defender second) requires opening a neighbor proactively, despite available non-neighbor opens, to turn the whole queue into an even-length odd-degree corridor; every old-menu move loses. Adding that corridor certificate is still incomplete: **GCZMmw** requires opening a non-neighbor and deliberately leaving an odd front unrepaired. These are the first exact

witnesses that the prevention/debt induction needs recursive, proactive debt rather than a parity-local trap trigger. The broader no-self-flip envelope—all opens and exactly the even-front closes in prevention mode—contains both winning moves; menu choice remains existential, not universal. Paired with the old debt menu, that envelope is strictly complete against an unrestricted attacker on all 12,346 $k = 8$ classes, both seats.

Open residue. The general- n proof. Parity-local invariants provably do not suffice: over 13 natural parity features of the state, the safe/unsafe label is not a function of the features, and minimal distinguishing pairs differ precisely in whether the untouched structure retains a repair device – a recursive, not parity-local, condition. The old parity-local R3/D3 realization of the firewall induction is now falsified by the two $k = 8$ menu witnesses, not merely unfinished. The replacement route combines block turns with live-degree pairing. In the scoring form above, a second-seat defender can mirror a close by the forced next close for free, and can mirror an open x by an untouched y of the same live-degree parity. This is a complete proof whenever the vertices admit a perfect matching into true/false twin pairs: twins retain equal live-degree parity in every live set containing them, opens and closes consume pairs together, and every two-move score is zero. In the general graph the required partner may already be queued; equivalently, the attacker uses paired front deletions to reshape the live degree classes. The hard obligation is now exact: prove that the isolated coin supplies or regenerates a same-parity partner (including the exceptional dummy-open/all-others-odd case), or find a graph where that repair debt cannot be discharged.

11 Resolution: the weighted-source Witt–FIFO game

The arbitrary-graph linking theorem above is stronger than the Gold problem requires. The polar form is public data, so it may be put into a deterministic Witt frame before play. This changes the strategic support graph from an arbitrary induced graph to a matching. A naive version would attach the dense value $Q(f_i)$ to a Witt-basis coin f_i , violating N2 in the original Grundy frame. The decisive point is to split that value into a public polar part and weight-one source *interactions*: the refinement bits weight local overlap edges rather than being accumulated as invariant terminal labels.

Fix the original frame $e_1, \dots, e_m$ and write

$$P_B(z) = \sum_{j < k} B(e_j, e_k) z_j z_k, \quad Q_q(z) = P_B(z) + \sum_j q_j z_j, \quad q_j = Q_q(e_j).$$

Choose, by a fixed Gaussian-elimination convention depending only on B , a symplectic-plus-radical basis

$$f_i = \sum_j C_{ji} e_j, \quad B(f_{a_h}, f_{b_h}) = 1$$

whose other pairings vanish. For $x = \sum_i y_i f_i$, put

$$p_i = P_B(f_i) = \sum_{j < k} C_{ji} C_{ki} B(e_j, e_k).$$

All of C , $y = C^{-1}x$, and p_i are refinement-blind public data.

For completeness, the adapted basis exists by the usual two-line induction for alternating forms. If $B = 0$, retain any basis as radical vectors. Otherwise choose the lexicographically first pair u, v with $B(u, v) = 1$. The plane $H = \langle u, v \rangle$ is nonsingular, since a vector $au + bv$ orthogonal

to both u and v has $a = b = 0$; hence $V = H \oplus H^\perp$. Recurse on $H^\perp$. Choosing the first available pivot and performing row operations in the fixed original-coordinate order makes the construction deterministic. It uses only the public matrix B , works unchanged in the singular case, and terminates after one dimension drop for a radical step or two for a hyperbolic-plane step.

Rule (weighted-source witt-ffo). Load a strategic coin F_i for each $y_i = 1$, with fixed public diagonal charge p_i . Join F_{a_h} to F_{b_h} exactly when both are loaded. Every loaded strategic Witt edge has weight one. For each original coordinate $x_j = 1$, load in addition a q-blind source pair S_j^0, S_j^1 . Its potential edge has local weight $q_j = O_q(e_j)$: opening one endpoint while its mate is already queued xors q_j into the charge. Source closes have charge zero.

Every loaded coin must be touched twice. An untouched coin may be opened; only the FIFO front may close; opening onto an empty queue sets the one-move ko; and a player passes exactly when no open or close is legal, clearing ko. Initialize $\sigma = 0$. When a coin v opens, update σ by the weight of its already-open mate, if any. Thus every strategic or source edge is charged exactly when its two touch intervals overlap. When a strategic front F_i closes, update

$$\sigma \oplus = p_i.$$

All other closes and all passes leave σ unchanged. After the queue is drained, add one ordinary normal-play claim move exactly when $\sigma = 1 \oplus \phi$, where ϕ is the current move-parity bit initialized to 0 and toggled after every move.

For a Gold form, Theorem 1 supplies $q_j = \text{Tr}((w_a^2 + w_a)e_j)$ for the source-pair weight. Definition 2 counts this as one weight-one oracle observation on an overlapping source OPEN. It does not claim that constructing w_a by exact finite-field linear algebra, or evaluating its trace, is itself one primitive game move.

The move-parity bit is not an evaluator or form oracle; it is already the mover field of the FIFO state. It is necessary for an identity-valued payoff compiler: a phase-free terminal leaf assigns opposite initial-seat winners at even and odd depths. The final claim is therefore a two-state local “charge lantern”, not an inspection of x , B , or $Q(x)$.

Theorem 7 (FIFO matching theorem). *Let every component of G be K_1 or K_2 . From the empty-queue root, either designated seat has a strategy forcing the sum of all close charges $\deg_U(f)$ to be zero. In fact the strategy makes every close charge zero individually. The theorem holds for every finite board and needs no dummy.*

Proof. Call a FIFO front *safe* when either ko is set or its unique mate is not untouched; equivalently, every legal close at that checkpoint has charge zero. Induct on $4|U| + 2|Q| + [\text{ko}]$. We show that a designated seat forces terminal score zero whenever it moves from score zero, and whenever its opponent receives a score-zero safe state.

If the designated seat moves with $U \neq \emptyset$ and the queue is empty, OPEN any $z \in U$; ko makes the next checkpoint safe. If the front f has charge one, its unique mate u lies in U ; OPEN u , which erases the only possible live neighbour of f . If f has charge zero, OPEN any $z \in U$; maximum degree one implies that erasure preserves the zero front. When $U = \emptyset$, all later closes have charge zero and the forced PASS, when present, is harmless.

At an opponent checkpoint, OPEN and PASS preserve the score and return the turn. A legal CLOSE rules out ko, so safety makes its charge zero and likewise returns a score-zero state. Every branch has smaller rank. Initially the queue is empty: the second seat receives a vacuously safe root,

while the first seat uses the empty-queue OPEN. The designated seat itself never closes while an untouched coin remains, and after the untouched set is empty every close has charge zero. Thus the constructed strategy makes each close charge zero, not only their total. This proves both seats. $\square$

The exact transition-system proof is kernel-checked as `evenWins_initial_of_matching` in `formal/0gdoad/FifoMatching.lean`. The finite matching census helped find the policy but is not used in the theorem. Its refinement-blind strengthening, with a public potential matching and an arbitrary edge-deleted scoring submatching, is kernel-checked as `evenWins_initial_of_every_submatching`. The proposition is pointwise in the scoring submatching, while its proof branches choose moves only from the public matching. Since `EvenWins` does not reify a first-class policy or complete play traces, the literal single-policy quantifier and the per-close-zero observation are the explicit argument above, not separate theorem statements in Lean.

Theorem 8 (weighted-source exactness). *For every $\mathbb{F}_2$ -valued quadratic refinement Q_q on a finite-dimensional $\mathbb{F}_2$ -space, every input x , and either charge stance, weighted-source WITT-FIFO forces terminal charge $\sigma = Q_q(x)$. Consequently its charge-lantern normal-play root is P if and only if $Q_q(x) = 0$.*

Proof. Quadratic expansion in the adapted basis gives

$$P_B(x) = \sum_i y_i P_B(f_i) + \sum_{i < k} y_i y_k B(f_i, f_k) = \sum_i y_i p_i + \sum_h y_{a_h} y_{b_h}.$$

The linear refinement part transports without any dense query:

$$\sum_i y_i \sum_j C_{ji} q_j = \sum_j x_j q_j.$$

Therefore

$$Q_q(x) = \sum_i y_i p_i + \sum_j x_j q_j + \sum_h y_{a_h} y_{b_h}. \quad (2)$$

Form instead the q -blind *potential matching* P : retain every loaded strategic Witt edge and every loaded source-pair edge, irrespective of its weight q_j . Apply the explicit safe-front strategy of Theorem 7 to P . The policy consults only this potential matching and the public FIFO state: when the front's potential mate is still untouched it opens that mate, and otherwise it opens any untouched coin. It is therefore the same strategy for every refinement q . The proof of the theorem shows more than cancellation of a total score: every close has zero live P -degree.

For one potential edge e , exactly one of the following occurs: its two touch intervals overlap, or the earlier-opened endpoint closes while the other endpoint is still untouched. The latter case would give that close live P -degree one, so the strategy excludes it. Consequently every potential pair overlaps. The rule's total OPEN charge is therefore directly

$$\sum_h y_{a_h} y_{b_h} + \sum_j x_j q_j,$$

with no refinement-dependent auxiliary graph or strategy. Every active strategic F_i closes exactly once and supplies $\sum_i y_i p_i$. Equation (2) therefore gives forced terminal charge $Q_q(x)$ at both stances.

For the final assertion, retain the finite charge game tree. At a leaf of charge σ and phase ϕ , the unique claim exists exactly when the player to move is the seat designated by σ . Backward induction

says that the compiled normal-play winner equals the charge-game winner at every subtree and phase. At the root, seat zero plays for charge one, so the root is N for $Q_q(x) = 1$ and P for $Q_q(x) = 0$. The general compiler and root equivalence are kernel-checked in `formal/0gdoad/GoldSemantics.lean`. $\square$

Proposition 5 (locality and optimal observation support). *The rule satisfies F1–F3 and N1–N2 with $(w_0, c) = (1, 1)$. Its distinct refinement observations at input x are exactly the singleton directions $\{e_j : x_j = 1\}$; they span x and attain the weight-one lower bound of Theorem 5.*

Proof. The basis algorithm, $x \mapsto y$, both loaded support sets, the matching potential edges, public charges p_i , and terminal board condition use only B, x and the fixed frame. Thus F1 holds. An OPEN involving a source pair needs at most the one singleton value $O_q(e_j)$, and only when its mate is already queued; every other move predicate and transition is refinement-blind. Even the forcing strategy is refinement-blind: it is the safe-front policy on the public potential matching P . This proves F2 and N2 with $(1, 1)$. The semantics is the single normal-play claim rule, giving F3, and Theorem 8 is N1 for every refinement of B .

There is one source pair exactly when $x_j = 1$, so the set of possible oracle directions is exactly $\{e_j : x_j = 1\}$. These vectors are independent and sum to x ; no proper subset spans x . Theorem 5 says every transcript-stable exact weight-one interface must cover the same support. $\square$

The oracle notation is accounting, not hidden information: q is a public parameter of the perfect-information game. The source-pair repair matters nevertheless. With the earlier invariant source closes, every complete history had the form

$$\sigma_q(h) = c_{B,x}(h) + \sum_{j:x_j=1} q_j,$$

so two refinements agreeing on $Q(x)$ produced action-labelled isomorphic game trees. In the present rule the coefficient of q_j is the play-dependent overlap of S_j^0, S_j^1 ; refinement data therefore enters the same local interaction law as the polar edges, not a history-independent terminal bit. This is an intensional design fact, not an outcome-only certification: Theorem 6 proves that the reachable and unavoidable fork properties do not provide such a certification.

The rank-zero boundary is left undecorated. The same weighted-source rule still realizes the linear refinement exactly; there simply are no strategic Witt edges to pretend otherwise.

The original isolated-dummy arbitrary-graph linking conjecture remains an interesting strict generalization, but it is no longer load-bearing for Gold: the public Witt frame reduces precisely the required boards to the proved matching class, while weighted source pairs preserve fixed-frame locality.

12 Status and remaining generalizations

Status of the trichotomy. Tier 1 ($\text{Sp}(B)$ -blind) is dead by Proposition 1 and Theorems D–G. Tier 3 (evaluation) is live and even decision-nondegenerate, but tautological. The direct-board bounded-framing quarantine contains only clocks at dense positions (Theorem 2). Weighted-source Witt–FIFO escapes that direct board by a q -blind structural lift. Its matching theorem is uniform in dimension, its transitions are singleton-local, and its claim lantern gives an ordinary normal-play P -set. It is not a bounded-total-observation rule: Theorem 5 proves that no torsor-uniform exact rule can be one. Instead it attains the necessary support span exactly and uses every refinement bit as a local overlap weight. This is the sharp objective boundary; Theorem 6 explains why these outcome-only fork screens supply no further theorem of “non-tautology”.

Remaining generalizations. The isolated-dummy arbitrary-graph linking theorem of Section 10.5 remains open, with the exact global affine-contraction frontier recorded in `writeups/linking_affine.tex`. It would strengthen FIFO combinatorics but no longer strengthens Gold exactness. The historical family sweeps (ko-window, alternate pass rules, pair touches), Frobenius-aware enumeration, conjugation-move census, and Plambeck–Siegel regularity check are likewise calibration or independent continuations, not premises of Theorem 8. The all-exponent diagonal source is supplied by Theorem 1; sourcing is no longer an open premise.

13 Validation

The principal game statement is proved structurally and its load-bearing parts are checked in Lean. The older implementation checks remain as independent regression evidence and historical provenance.

- `formal/Ogdoad/FifoMatching.lean` kernel-checks the exact transition-system proof of the both-seat matching strategy and its public- matching construction parameterized over every edge-deleted submatching. The q-blind single-policy and per-close trace observations use the visible move branches of that construction but are not separately reified Lean predicates. `formal/Ogdoad/GoldMatchingAlgebra.lean` checks the abstract adapted-support expansion and split-diagonal identity from their supplied Witt-pairing and diagonal hypotheses; the concrete matrix C and $p_i = P_B(f_i)$ instantiation is the elementary polarization calculation in the paper, not an encoded basis algorithm. `formal/Ogdoad/GoldSemantics.lean` checks the recursive winner equivalence for the claim compiler; it does not construct a second explicit move-graph datatype. `formal/Ogdoad/GoldNoEvaluator.lean` checks the span/support observation lower bounds, and `formal/Ogdoad/GoldForkPadding.lean` checks outcome-preserving unavoidable fork padding. `formal/Ogdoad/GoldDiagonal.lean` checks the quadratic-tower trace reconstruction, absolute-trace descent, and finite-field Artin–Schreier exactness behind the all-exponent source. The paper supplies the joins between these independently checked lemmas; there is not a single end-to-end Lean theorem constructing the whole weighted arena. All six files contain no `sorry`, `admit`, or custom axioms.
- `cargo test` exercises the scalar arithmetic, Clifford product, Arf computation, quadratic fitting, misere machinery, and related invariant modules.
- `experiments/trace_form_arf.py` checks the Gold rank formula (Table 1).
- `experiments/gold_form_from_games.py` and `experiments/tartan_bilinear.py` rebuild the Gold form and polar form from literal Turning-Corners products in small fields.
- `experiments/arf_win_bias.py` checks the zero-count formula (Table 2).
- The `misere_quotient`, `octal_hunt`, `interactive_kernel`, `loopy_quadric`, and `bent_route` examples implement the route probes of Section 6.1.
- `experiments/gold_family_survey.py` checks the scaled-component bent counts; `experiments/framing_obstruction.py` the diagonal-framing pattern; `experiments/misere_kernel.py` the R_8 kernel reading.
- `experiments/gold/` carries the probes of the 2026-06-10 parallel run backing Sections 8–10: exhaustive symmetry checks at $m = 4$, the B -oracle and flip-rule kills, the extraspecial and

ECHO solvers, and the criterion calibration. These are machine-generated research probes, run under the project virtualenv, and are *not* CI-maintained; each result cited from them states its verification scope inline above.

- `experiments/echo_solver.py` is the adversarial-review harness of the 2026-06-10 verification pass (stages `selftest`, `ko2`, `fifo2-validate` through `fifo2-all`): direct full-state solvers for the ECHO family, self-tested against the Turning-Corners mex recurrence, explicit tree enumeration, and the original direct solver run verbatim. Unlike `experiments/gold/` it is dependency-free (stdlib Python) and written to stay reproducible.
- `experiments/linking_game.py` is the linking-reduction harness of the second 2026-06-10 pass (stages `validate`, `screen`, `strategy`, `graph6`; Section 10.5): the abstract odd-close parity game, the reduction identities on random legal plays, the all-classes $k \leq 8$ rigidity and Bad-graph screens, the two first menu counterexamples, and strict verification of the no-self-flip/debt envelope. Stdlib-only; cross-validated against `experiments/echo_solver.py` through its `SynthForm` bridge.

Two probe-hygiene caveats from the run remain open in the repository and are recorded in Remark 9: the `octal_hunt` target should be reframed (its group-quotient target is empty under Theorem C), and `p_set_as_f2` should verify that its subset labeling is a homomorphism.

14 Conclusion

The meaningful content is layered. At the base: game-operation-built characteristic-2 trace forms inside a nimber Clifford backend, with Arf and the singular radical data now determining the P -set bias of weighted-source WITT-FIFO. Above it: the linear ceiling (Theorem A, with the lexicodes as its rich solved floor), the extraspecial identification of the missing quadratic datum with a noncommutative central extension (Lemmas 1–3), and a no-go program (Theorems B–H, 2–3) that empties every symmetric, commutative, and oracle-bounded architecture of bulk strategic content. At the historical boundary: exact direct-board normal-play realizers in the tested quarantine are forced clocks; ECHO-ko is exact on the $m = 4$ bent instances at one stance, degrades gracefully with polar rank, and stays decision-live; and the ECHO-FIFO+dummy realizer is now *verified* exact on all 765 scaled $m = 8$ Gold forms at both stances – decision-nondegenerate, bounded-access, and σ -valued (Section 10.3). The resolution uses a q-blind structural lift outside that direct-board quarantine: a Witt frame makes the polar graph a matching, weighted source pairs preserve original-frame singleton locality, the both-seat safe-front theorem forces the exact charge in all dimensions, and the phase-aware claim lantern compiles it to normal play. Exactness itself forces every stable observation certificate to span the input, so the rule’s support-sized singleton access is optimal; and unavoidable fork padding proves that the reachable/optimal/unavoidable fork properties cannot certify naturality. The remaining arbitrary-graph FIFO contraction is a strict combinatorial generalization, not a gap in the Gold theorem.

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